

Data Types

A Data Type tells the compiler what type of value a variable can store. In simple words, before storing any value in a variable, we must tell Java what kind of value will be stored.

Example

```
int age = 25;
```

Here:

- int is the data type.
- 25 is an integer value.

Types of Data Types

Java data types are mainly divided into two categories:

1. Primitive Data Types

These are predefined data types provided by Java.

- byte
- short
- int
- long
- float
- double
- char
- boolean

2. Non-Primitive Data Types

These are created by programmers or provided by Java.

Examples:

- String
- Array
- Class
- Interface

Primitive Data Types

PRIMITIVE VALUES		PRIMITIVE DATA TYPES	SIZE
Numbers	Integer (Whole number) +ve to 0 to -ve	byte	1 byte
		short	2 byte
		int	4 byte
		long	8 byte
	Floating value	float	4 byte
		double	8 byte
Character		char	2 byte
Boolean		boolean	Undefined

1. byte

Used to store small integer values.

```
byte age = 25;
```

```
System.out.println(age);
```

2. short

Used to store integer values larger than byte.

```
short marks = 500;
```

```
System.out.println(marks);
```

3. int

Most commonly used data type for storing whole numbers.

Example

```
int number = 12345;
```

```
System.out.println(number);
```

4. long

Used to store very large integer values.

```
long population = 8000000000L;
```

```
System.out.println(population);
```

5. float

Used to store decimal values.

```
float percentage = 85.5f;
```

```
System.out.println(percentage);
```

6. double

Used to store large decimal values with better precision.

```
double price = 999.99;  
System.out.println(price);
```

7. char

Used to store a single character.

```
char grade = 'A';  
System.out.println(grade);
```

8. boolean

Used to store true or false values.

```
boolean result = true;  
System.out.println(result);
```

Variables in Java

A variable is used to store data in a program. It acts like a container that holds a value.

Every variable has a name so that we can access and use its value whenever needed. The value stored in a variable can be changed during program execution.

Example

```
int age = 25;
```

Here:

- int → Data Type
- age → Variable Name
- 25 → Value

Syntax of Variable

```
dataType variableName = value;
```

Example

```
int age = 25;
```

Creating Different Variables

Example

```
public class Demo {  
    public static void main(String[] args) {  
        int age = 25;
```

```
double salary = 45000.75;
char grade = 'A';
boolean passed = true;
String name = "Rahul";
System.out.println(age);
System.out.println(salary);
System.out.println(grade);
System.out.println(passed);
System.out.println(name);
}
}
```

Output

25

45000.75

A

true

Rahul

Rules for Naming Variables in Java

A variable name should follow some rules defined by Java. If these rules are not followed, the program will give compile time error.

Rule 1: Variable Name Cannot Start with a Number

Correct

```
int age1 = 25;
```

```
int student123 = 100;
```

Incorrect

```
int 1age = 25;
```

```
int 123student = 100;
```

Rule 2: Variable Name Can Start with a Letter or _ (Underscore) or \$ (Dollar Sign)

Example

```
int age = 25;
```

```
int _count = 10;
```

```
int $id =12345;
```

Rule 5: Spaces Are Not Allowed

Incorrect

```
int student age = 25;
```

Correct

```
int studentAge = 25;
```

Rule 6: Special Characters Are Not Allowed

Incorrect

```
int student-age = 25;
```

```
int age@ = 20;
```

Rule 7: Java Keywords Cannot Be Used as Variable Names

Incorrect

```
int class = 10;
```

```
int public = 20;
```

Correct

```
int classNumber = 10;
```

```
int publicCount = 20;
```

Rule 8: Variable Names Are Case Sensitive

Example

```
int age = 25;
```

```
int Age = 30;
```

Both are different variables.

Rule 9: Variable Name Should Be Meaningful

Poor Naming

```
int a = 25;
```

```
int i =12345;
```

Better Naming

```
int age = 25;
```

```
int id =12345;
```

****Meaningful names make code easy to understand.**

Use meaningful names.

Examples

```
int age = 25;
```

```
double salary = 45000.50;
```

```
char grade = 'A';
```

```
boolean isPassed = true;
```